

Duc Nghiem

Last updated in June 2026

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Computer Science student at UMass Amherst seeking a Software Engineering/AI Engineering/AI Research Assistant/IT opportunity, with interests in building scalable systems and applying machine learning to real-world problems.

EDUCATION

University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Computer Science

09/2025 – 12/2028

- Class Standing: Sophomore.
- GPA: 3.786/4.00.
- Achievements: Dean's List.

RESEARCH PAPERS AND POSTERS

- Mihir Chandra, Yijie Zhou, Chase Borges, Anna Chatterji, Ilia N Ivanov, **Duc Nghiem**, et al. (2026). "Probing Charge-Carrier Doping Effects on Interchain Thermal Transport in Edge-On-Oriented Conjugated Polymer Thin Films." *Surfaces and Interfaces (Q1 journal)*. [Link] 📄
- Henry Worden, Mihir Chandra, Yijie Zhou, Zarif Ahmad Razin Bhuiyan, Mouyang Cheng, Krishnamurthy Munusamy, **Duc Nghiem**, et al. (2026). "Suppressing Thermal Transport in Nonporous Polymer Hybrids by Limiting Thermally Accessible Vibrational Modes." *Materials Horizons (Q1 journal)*. [Link] 📄
- **MassURC Conference Presenter**: Designed, presented the abstract and poster "Intermittent Machine Learning for Battery-Free Environmental Sensing Under Solar Power" alongside over 1,000 presenters and visitors. [Link] 📄

INTERNSHIP

Viettel AI 📄

Hanoi, Vietnam

AI Engineer

05/2026 – 08/2026

- Built FastGraphRAG/LightRAG/Microsoft GraphRAG backend services with Neo4j for Vietnamese government legal documents used by thousands of users, developing the end-to-end pipeline from PDF processing with Marker to improve the workflow and entity extraction to knowledge graph construction and multi-hop RAG retrieval via APIs.
- Built an MCP server for Viettel Agent to generate legal-document videos through HyperFrame and developed a Chatwoot-based Facebook chatbot, integrating AI-powered legal information retrieval and automated user interactions, reducing the need for human intervention.

RESEARCH EXPERIENCE

SBU Graph Lab - Professor Tengfei Ma 📄

Remote, NY

Student Researcher

05/2026 – Present

- Built DimeNet+ and MACE from scratch for molecular energy prediction, training on AQM dataset and fine-tuning on FreeSolv with ASE-based atomistic workflows
- Developed uncertainty-aware prediction pipelines with deep ensembles and systematic benchmarking, improving model reliability and validating performance against leading FreeSolv baselines; achieved a best MAE of 0.418, matching state-of-the-art FreeSolv performance.

Advanced Polymer Engineering Laboratory - Professor Yanfei Xu 📄

Amherst, MA

Research Assistant

10/2025 – Present

- Developed Python/PyTorch algorithms for polymer thermal conductivity analysis, deriving material parameters and uncertainty estimates; contributed to data curation, formal analysis, validation, and research methodology.
- Curated and analyzed large-scale polymer/crystalline material datasets using Pandas/PyTorch and quantified pore areas with ImageJ, supporting materials characterization and publication-ready research.

Wireless and Sensor Systems Lab - Professor VP Nguyen 📄

Amherst, MA

Research Assistant

09/2025 – 08/2026

- Engineered a battery-free, solar-powered imaging system website with an automated upload, integrating energy harvesting, on-device image capture, and cloud infrastructure for extended field deployment.
- Optimized YOLOv8n-Nano/NanoDet-Plus for real-time object detection at ~34 FPS on resource-constrained Android devices and developed PyTorch diffusion/GAN pipelines with Gemini API for high-resolution image restoration used in research publications.

SOCIAL ACTIVITIES

BUILD UMass 📄

Amherst, MA

Software Developer / Interviewer

09/2025 – 05/2026

- Worked on reducing emergency-response latency for healthcare users by an estimated ~32–51% by architecting and deploying a scalable backend system for the Kapok mobile application.
- Engineered real-time routing and connection systems using the MapBox API, built the database with Firebase, and designed the frontend with Figma and Flutter, enhancing communication and data management.
- Interviewed software development candidates for the BUILD UMass Spring '26 season using structured technical assessments and behavioral interviews, resulting in the selection of a high-performing cohort that met project staffing goals.

PROJECTS

LLM Guardrail | [Try it out](#) | [Github](#)

04/2026 – 06/2026

- Built a multi-tenant LLM Safety Gateway — designed and implemented a production-grade FastAPI API gateway that sits between applications and LLM providers (OpenAI, Anthropic, Gemini, Groq, Ollama), enforcing per-organization guardrail policies (prompt injection detection, jailbreak prevention, PII redaction/blocking, secret leakage detection, toxic content filtering, and topic-based compliance rules) with Redis-backed rate limiting, full audit logging (PostgreSQL), and JWT + API key authentication.
- Integrated multiple LLM providers, including Groq, OpenAI-compatible APIs, Anthropic, Gemini, Ollama, and mock backends, through a modular provider layer for scalable model routing and testing.
- Hardened the project for public release with Dockerized deployment, environment-based secret management, security headers, ignored credential files, automated migrations, and backend tests.

Drag Soccer with Agent | [Try it out](#) | [Github](#)

08/2025 – Present

- Built a full-stack multiplayer soccer simulation platform (Flask + Redis + Supabase) featuring real-time 1v1 online play via invite links, friend system, and single-elimination tournaments with auto-advancing AI matches, deployed on Render with 7 built-in AI opponents.
- Created an Agent Skill Vulnerability Scanner — developed a static-analysis engine that scans Cursor skills, MCP instructions, and system prompts for hard-coded secrets (API keys, tokens, credentials), PII, database URLs, private IPs, internal file paths, and 16 destructive shell/SQL command patterns (rm -rf, DROP TABLE, disk wipe, fork bombs, curl-pipe-shell), complete with a web review queue for managing blocked findings, per-user override allow-lists, and git pre-push hooks to prevent leaking sensitive data to GitHub.
- Delivered a Full-Stack Platform with Token-Based Billing — built a React SPA dashboard (14 views including playground, analytics, billing, admin, logs, and policy management) alongside a Stripe-integrated token metering system that tracks per-request LLM token consumption across four paid tiers (9–399), with automated low-balance alerts, unlimited-token whitelisting, demo mode with strict rate caps, and Docker Compose deployment ready for Render, Koyeb, or bare-metal hosting.

HONORS & AWARDS

- **Best Use of DigitalOcean** @ Hack(H)er413 2026 – Built ShareBooth, a browser-based real-time photobooth enabling users worldwide to capture and compose group photos on a shared canvas; implemented client-side photo compositing with the Canvas API and background removal using MediaPipe Selfie Segmentation (~30fps), and deployed a Node.js/Express backend on DigitalOcean for session management and WebSocket coordination. [\[Link\]](#)
- **Third Place** @ MassAI's Winter Break Challenge – Developed a 93.8% accuracy spam detection model by fine-tuning a BERT model using PyTorch, Python, Pandas, Scikit-Learn, NumPy, Matplotlib, T4 GPU, and Google Colab. [\[Link\]](#)

CERTIFICATES

- **Software Engineering Job Simulation** @ Hewlett Packard Enterprise [\[Link\]](#)
- **Advanced Software Engineering Job Simulation** @ Walmart Global Tech [\[Link\]](#)

SKILLS

- **Languages:** Python, Java, HTML, CSS, JavaScript, SQL, LaTeX, R
- **Frameworks:** React, Node.js, Flask, FastAPI, WordPress, AutoCAD, Flutter
- **Libraries:** Pandas, PyTorch, TensorFlow, NumPy, Matplotlib, OpenCV
- **Developer Tools:** Git, GitHub, Google Cloud Platform (GCP), AWS, Docker, VS Code, PyCharm, Vim, Windows Terminal, Neo4j, Render, Redis, Stripe, Microsoft Word, Excel, Powerpoint, Microsoft 365, Kubernetes, Origin, ImageJ